Introduction To Computational Biology

Sem 1 | Watermarked study resource | Generated from local SRM dataset

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===== SOURCE_FILE: nexus sem/Introduction To Computational Biology/PYQs/HOTs Questions Unit 1 to 5, Question Banks, MCQs & More.url =====
CATEGORY: PYQs
STATUS: url_reference_only

URL: https://drive.google.com/drive/folders/1wN2x025nxZYsCoUhyNoHJUM8WT1i-sWT?usp=drive_link

===== SOURCE_FILE: nexus sem/Introduction To Computational Biology/PYQs/PYQ Jan 2023.pdf =====
CATEGORY: PYQs
STATUS: ok

[Page 1]
Note: (i) (ii)
B.Tech. / M.Tech (Integrated) DEGREE EXAMINATION, JANUARY 2023 First Semester
Time: 3 Hours
21BTB102T--INTRODUCTION TO COMPUTATIONAL BIOLOGY T'or the candidates admitted from the acadenic year 2022-2023) Part -A should be answercd in OMR shcet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minulc. Part-B and Part -C should be answered in answer booklet.
1. In cell division, (A) Cytokinesis
Page I of 3
(C) Interphase
{A) Nucleus (C) Nucleoid 2. The genetic material of a prokaryotic is present in the
(C) Schwann
Reg. No.
3. The term "Cell" was given by A) Robert Hooke
4. Ribosomes are also known as (A) Power house (C) Protein factory
(A) tRNA
PART-A (20 x 1 = 20Marks) Answer ALL Qucstions
(C) DNA 6. Blastocyst is
is for growth and maintenance. (B) Meiosis (D) Mitosis
(A) Totipotent cell (C) Zygote
5. Genetic information is stored in the formed of
sequence database (A)_BLASTn (e) BLASTx
(B) Cytoplasm
(C) Vander walls interaction

(D) Plasmid
(B) Jatum (D) Debary
(B) Store house (D) Suicide bag
(B) mRNA (D) RNA
7. DNA is made of hydrogen bands they are (A) lonic bond
(B) Embryo D) Early stage of embryo
(B) Covalent bond
8. A program compares amino acid query sequence against a protein (D) Non bonded interaction
(B) BLASTp (D) TBLASTX
Max. Marks: 75
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(B) GGA 9. Protein synthesis stops when encountering stop codon (D) CGA (A) AUG (C) UGA (B) Ribosome 10. The amino acids are assembled into a polypeptide in (A) mRNA (D) RNA (C) DNA
11. Non-proteinaceous molecule needed for enzyme activity is called
(A) Cofactors
12. Pymol is a (C Hormones
(A) Visualization tool (C) Python tool
17.
(A) Microglia
14. Synaptic signaling involves (A) Autocrine signals (C) Neurotransmitters
(C Astrocytes 13. Which of the following cells have star shaped structure?
Page 2 of 3
16. Parkinson's is caused by
(A) Soma (C) Dendrite
A) Amyloid plaques
15. The connecting weights in a ANN is equivalent to
(C) Dopamine related neurons
(B) Enzymes
19. Tc cells are
(D) Inhibitors
{B) Statistic tool
(A) Cytotoxic T cells
(D) Sequence analysis tool
(C) Ciliated T cells
(B) Digodendrocytes (D) Schwann cells
(A) Linear (C) Long
(B) Paracrine signals (D) Neurochannel signals
(B) Axon
is a plasma-like liquid carried by lymphatic circulation. B) Blood (A) Lymph (C) Saliva 18. Polio drops are administrated at infant stage because (A) It increases RBC count
D) Synapse (B) Neurofibrillary tangles (D) Spongiform
(D) Gastric juices
(C) Corrects bone marrow stem (D Produces antibodies cells
of neurOn.
(B) Helps better digestion
20. Epitope prediction of B cell is based on
antigens
(B) Cellular T cells (D} Cell mediatedT cells

(B) Conformational D) Core of antigen
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PART -B (4 × 10 = 40 Marks) Answer ANY FOUR Questions 21. Give the different organelles of cells. Briefly its structure and function. 22. Describe what is DNA and RNA? 23. Write the types of secondary structure prediction tools. 24. Describe machine learning and its uses in biology. 25. Ilustrate with neat diagrams cell mediated immune response. 26. Detailed overview of stem cell technology. PART-C(1x15 =15 Marks) Answer ANY ONE Questions 27. Give the algorithm to predict similar sequences to a protein sequence using a database.
28. What are the immune processes that happen when we got vaccinated for COVID?
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===== SOURCE_FILE: nexus sem/Introduction To Computational Biology/PYQs/PYQ Jul 2023.pdf
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CATEGORY: PYQs
STATUS: ok

[Page 1]
Page 1 of 3 20JF1/2-21BTB102T

Reg. No.

B.Tech. DEGREE EXAMINATION, JULY 2023

First / Second Semester

21BTB102T – INTRODUCTION TO COMPUTATIONAL BIOLOGY
(For the candidates admitted from the academic year 2022-2023)
Note:
(i) Part - A should be answered in OMR sheet within first 40 minutes and OM R sheet should be handed
over to hall invigilator at the end of 40th minute.
(ii) Part – B and Part – C should be answered in answer booklet.

Time: 3 Hours Max. Marks: 75

PART – A (20 × 1 = 20Marks)
Answer ALL Questions
Marks BL CO PO
1. Which one is involved in photosynthesis? 1 1 1 4
 (A) Chloroplasts (B) Endoplasmic reticulum
 (C) Cell wall (D) Ribosomes

2. Name the process where paternal and maternal genetic materia l is
exchanged
1 1 1 4
 (A) Bivalency (B) Crossing over
 (C) Synapsis (D) Dyad formation

3. An example of single celled eukaryotic organism. 1 1 1 4
 (A) Yeart (B) Ferns
 (C) Bacteria (D) Protozoa

4. The organelle that has fluid filled sacs for storage 1 1 1 4
 (A) ER (B) Golgi
 (C) Vacuole (D) Centriole

5. What is the lipid related disease? 1 1 2 1
 (A) Scurvy (B) Rickets
 (C) Sickle cell anemia (D) Atherosclerosis

6. Carbohydrates are poly hydroxyl complex of 1 1 2 1
 (A) Glycosidic bonds (B) Glucose and glucose
 (C) Glucose and alcohol (D) Aldehydes and ketones

7. DNA contains 1 1 2 1
(A) Glycosydic bond (B) Phosphodiester bond
(C) Iconic bond (D) Peptide bond

8. Where is lipopolysaccharide present? 1 1 2 1
(A) Cell cytoplasm (B) Cell membrane
(C) Cell junction (D) Cell

9. Which is not a stop codon? 1 1 3 2
(A) UAA (B) UAG
(C) UGA (D) CGA

10. Fibrous proteins have 1 1 3 2
(A) Structural functions (B) Metabolic functions
(C) Transport functions (D) Enzymatic functions

11. Amyloidosis is caused by 1 1 3 2
(A) Changes in secondary structure (B) Changes in tertiary st ructure
(C) Changes in quaternary structure (D) Changes in primary st ructure

12. PDB database is a 1 1 3 2
(A) Primary database (B) Secondary database
(C) Tertiary database (D) Composite database

13. The multiple projections from the cell body of neurons are 1 1 4 1
(A) Axons (B) Dendrites
(C) Spindle fiber (D) Flagella

14. Which of the following cells in the brain produced from mon ocytes? 1 1 4 3
(A) Microglia (B) Astrocytes

(C) Schwann (D) Oligodendrocytes

15. Multiple sclerosis is a disease caused by 1 1 4 1
(A) Schwann cells (B) Astrocytes
(C) Skin cells (D) Neuronal cell body

16. Type of machine learning are 1 1 4 1
(A) Deep learning (B) Supervised
(C) Unsupervised (D) Dynamic

17. Immune cells that allows for subsequent recognition of anti gen 1 1 5 1
(A) Memory cell (B) Plasma cell
(C) Basophil (D) APC

18. Name the process a cell such as a neutrophil or a macrophag e use to eat prey
1 1 5 1
(A) Chemotoxis (B) Phagocytosis
(C) Endocytosis (D) Exocytosis

19. Which is not an auto immune diseases? 1 1 5 1
(A) Influenza (B) Rheumatoid arthritis
(C) Multiple sclerosis (D) Lupers

20. ______ secrete histamines to attract other immune cells. 1 1 5 1
(A) Macrophages (B) Mast cells
(C) Platelets (D) Neutrophils

PART – B (4 × 10 = 40 Marks)
Answer ANY FOUR Questions

Marks

BL

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PO

21. State the properties and applications of stem cells.

10 2 1 4

22. Write a note on BLAST and its uses.

10 2 2 1

23. Describe transcription and the processing of its end product.

10 2 3 2

24. Write the basis of artificial neural network – also descri b e t h e p a r t s o f neural network.

10 2 4 1

25. Illustrate with neat diagrams of humoral immunity.

10 2 5 1

26.

Explain the types and uses of sequence databases. 10 2 2 1

PART – C (1 × 15 = 15 Marks)

Answer ANY ONE Questions

Marks

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PO
27. How a skin cell is different from a nerve cells?

15 4 1 4
28. If two protein sequences are similar - Explain (Justify)
(i) How their function will be similar?
(ii) The organisms that contain these proteins are evolutionarily
related

15 4 2 1

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===== SOURCE_FILE: nexus sem/Introduction To Computational Biology/PYQs/pyq txt/PYQ Jan 2023.txt =====
CATEGORY: PYQs
STATUS: ok

Reg. No.

B.Tech. /M.Tech (Integrated) DEGREE EXAMINATION, JANUARY 2023
First Semester

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T'or the candidates admitted from the acadenic year 2022-2023)

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PART-A (20 x 1 =20Marks) Marks BL CO PO

Answer ALL Qucstions
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(A) Cytokinesis (B) Meiosis
(C) Interphase (D) Mitosis

2. The genetic material of a prokaryotic is present in the 1 1

{A) Nucleus (B) Cytoplasm
(C) Nucleoid (D) Plasmid

3. The term "Cell" was given by 1 1 1 4

A) Robert Hooke (B) Jatum
(C) Schwann (D) Debary

4. Ribosomes are also known as 4

(A) Power house (B) Store house
(C) Protein factory (D) Suicide bag

5. Genetic information is stored in the formed of 1 2

(A) tRNA (B) mRNA
(C) DNA (D) RNA

6. Blastocyst is 1 1 2 1

(A) Totipotent cell (B) Embryo
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7. DNA is made of hydrogen bands they are 1 2

(A) lonic bond (B) Covalent bond
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8. A program compares amino acid query sequence against a protein 1 2

sequence database
(A)_BLASTn (B) BLASTp
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(B) GGA

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(A) Train

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PART – C (1 × 15 = 15 Marks) Marks BL CO PO

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(i) How their function will be similar?
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